StockEvent Lens — Batch 007 Intelligence Dashboard

Articles 031–035 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_07 (Articles 031-035)`

TEST 031
Original 100% Base
NVIDIA board approves $150B buyback plan, lifting total capacity to $235B[cite: 57, 59].

TEST 032
Repeat 96% Overlap
Re-syndicated wire report on $150B repurchase program with zero new facts[cite: 57, 60].

TEST 033
Update +22% New Info
Accelerates execution of first repurchase tranche, pushing shares past $240[cite: 57, 61].

TEST 034
Conflict Dispute (Rejection)
Contradicts buyback approval: claims board rejected plan, stock tumbles 4.5%[cite: 57, 62].

TEST 035
Different 0% Overlap
Unrelated announcement of Quantum-Accelerated Networking starting Group 014[cite: 57, 58].

Technical Analysis: Dense Vector Blindness vs. Numerical Contradiction Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *numerical or directional fidelity*. Changing corporate decisions (e.g., $150 billion approval vs complete plan rejection in test_034) yields high vector similarity (~0.89) because both texts share identical surrounding context tokens ("Nvidia", "board", "share repurchase", "$150 billion").

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard numerical or directional negation.
```

1. Dense Embedding Blindness
Dense embeddings project words into high-dimensional space where "approved $150B buyback" and "rejected repurchase plans" share near-identical context embeddings. Thus, pure vector cosine distance fails to flag this corporate decision conflict.

2. LLM Evaluation Override
To prevent missing critical governance disputes, the LLM reasoning layer performs explicit entity-attribute-value matching, overriding high vector similarity when $\Delta F_{contradiction} = 1.0$ is detected.

Article Relationship & Percentage Breakdown (Batch 007)

FILE IDENTIFIER	PATTERN LABEL	KEYWORDS IDENTIFIED	SIMILARITY %	DELTA %	CLASSIFICATION RATIONALE
`test_031_original`	Original (%5=1)	NVIDIA, $150B buyback, $235B capacity, all-time high, board approved[cite: 57, 59]	100%	Baseline	Establishes breaking story frame on NVIDIA's board approving a record $150B buyback[cite: 57].
`test_032_repeat`	Repeat (%5=2)	$150B repurchase, $235B capacity, 4.2% share reduction, EPS impact[cite: 57, 60]	96%	0% New Facts	Rephrased wire report detailing identical buyback authorization without new progress[cite: 57].
`test_033_update`	Update (%5=3)	**First tranche executed**, after-hours $240 breakout, institutional volume[cite: 57, 61]	84%	+22% Operational Facts	Advances event state to active execution of the first repurchase tranche[cite: 57].
`test_034_conflict`	Conflict (%5=4)	**Board rejected buyback**, capacity frozen at zero, stock tumbled 4.5%[cite: 57, 62]	89% Match	$\Delta F_{contradiction}$ = 1.0	Directly disputes buyback approval, claiming board rejected plan to fund manufacturing[cite: 57].
`test_035_different`	Different (%5=0)	Quantum-accelerated networking, Tokyo keynote, photonics latency reduction[cite: 57, 58]	0%	Group 014 Trigger	Zero domain overlap; triggers standalone Group 014 event cluster[cite: 57].

Group 013 — Consolidated Event Summary

4 Articles (031–034)

NVIDIA's board approved an additional $150 billion share repurchase authorization, lifting remaining buyback capacity to $235 billion through fiscal 2028 (Articles 031–032)[cite: 57]. Execution accelerated as NVIDIA **executed its first repurchase tranche early, driving shares past $240 in after-hours trading** (Article 033)[cite: 57]. However, market reports are in direct conflict regarding board approval: initial reports cited board approval of a $150 billion buyback lifting shares to record highs, whereas a contradictory report claims the **board rejected buyback plans completely to shift cash into manufacturing, causing stock to tumble 4.5%** (Article 034 ; $\Delta F_{contradiction}$ = 1.0)[cite: 57].

✓ Update Preserved: First Repurchase Tranche Executed
✓ Conflict Preserved: Board Approval vs Rejection ✓ Duplicate Removed: Article 032 Consolidated

Group 014 — Standalone Summary

1 Article (035)

NVIDIA unveiled its next-generation quantum-accelerated networking architecture in Tokyo, integrating hyper-scale photonics to cut data center latency by over 70%, with cloud deployments planned for early next year (Article 035)[cite: 57].

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 013

Batch 007 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

`test_031_original.json` — Group 013

```
{
  "id": 62142510,
  "headline": "QUICK SPARK: Nvidia Board OKs Massive $:
  "symbols": ["NVDA"],
  "summary": "Nvidia approved a massive $150B buyback,
}
```

Establishes $150B buyback approval baseline.

`test_033_update.json` — Group 013

```
{
  "id": 990000031,
  "headline": "Nvidia Accelerates Execution of Record $
  "symbols": ["NVDA"],
  "summary": "Nvidia rapidly deploys its newly approve
}
```

Adds first tranche execution & $240 breakout.

`test_034_conflict.json` — Group 013

```
{
  "id": 990000032,
  "headline": "Nvidia Board Rejects Buyback Expansion I
  "symbols": ["NVDA"],
  "summary": "Nvidia stock drops 4.5% as board denies a
}
```

Disputes buyback approval vs board rejection.